

Kareem Hassan

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EDUCATION

Purdue University

B.S. in Computer Engineering (Senior), GPA: 3.89

Concentrations: Artificial Intelligence & Machine Learning, Computer Systems

United States

2025 – 2026

Egypt University of Informatics

B.S. in Computer Engineering, GPA: 3.54

Egypt

2021 – 2025

TECHNICAL PROJECTS

Autonomous Navigation in Extreme Conditions (Senior Design) | Computer Vision, Sensor Fusion, Python 2026

- Designed a vision-based autonomous navigation system robust to adverse conditions such as rain, low visibility, and dynamic obstacles.
- Reduced reliance on traditional sensors by implementing real-time image-based perception and segmentation techniques.
- Integrated multi-sensor inputs and optimized processing pipelines for real-time performance.
- Collaborated within a multidisciplinary team to validate system performance in simulated environments.

Smart Cycle Helmet – Embedded Safety System (Team) | Python, ESP32, IMU, BLE, Machine Learning 2025

- Developed a real-time machine learning pipeline for crash, fall, and braking detection using IMU sensor data.
- Engineered feature extraction (acceleration, gyroscope, jerk) and trained a Random Forest model achieving 91% accuracy and 100% crash detection recall in real-time scenarios.
- Optimized inference for ESP32 deployment, achieving millisecond-level classification latency.
- Collected and processed 600+ event samples and integrated BLE communication for SOS alert transmission.

Reinforcement Learning with Human Feedback (RLHF) | Python, Machine Learning

2025

- Implemented a reinforcement learning pipeline incorporating human feedback signals to improve model behavior.
- Designed reward modeling strategies to align outputs with user preferences.
- Evaluated performance through iterative feedback loops and policy optimization techniques.

VAE-GAN Face Synthesis with U-Net Discriminator | Python, VAE, GAN

2025

- Implemented a hybrid VAE-GAN architecture for realistic face generation.
- Designed a U-Net discriminator with skip connections for multi-scale feature learning.
- Trained on the CelebA dataset, balancing adversarial and reconstruction losses for high-quality outputs.

EXPERIENCE

Team Member – Chick-fil-A

West Lafayette, IN, 2025 – Present

Customer Operations & Team Coordination

- Delivered high-volume customer service in a fast-paced environment while maintaining accuracy and efficiency.
- Collaborated with team members to optimize workflow and reduce service time during peak hours.
- Developed strong leadership, communication, teamwork, and problem-solving skills in real-time operational settings.

Intern – Orascom Construction

Egypt, 2024

Engineering & IT Department

- Assisted IT team with system maintenance, troubleshooting, and network reliability.
- Gained exposure to large-scale construction operations and IT integration in field environments.

Teaching Assistant (Mathematics & Chemistry)

2018 – 2023

Pre-College & College Level

Egypt

- Tutored students in mathematics and chemistry, improving average test performance by 15%.
- Designed problem sets and mock exams to reinforce conceptual understanding.
- Built leadership and communication skills through one-on-one and group mentoring.

TECHNICAL SKILLS

Languages: C, C++, Python, SQL

Embedded Systems: ESP32, IMU Integration, Real-time Systems

Machine Learning: scikit-learn, pandas, NumPy, feature engineering, model training

Tools: Git, Jupyter Notebook, LaTeX

Web Development: PHP, MySQL

Languages (Spoken): English (Fluent), Arabic (Native)

Other: Leadership, teamwork, communication, problem-solving

CERTIFICATIONS & HONORS

Purdue University: Dean's List & Honors List (Fall 2025)

Cisco Networking Academy: Networking Essentials (2024)

IBM SkillsBuild: Cybersecurity Certificate (2024)

Information Technology Institute (ITI): Full Stack Web Development using Python (2024)

Orascom Construction: Health and Safety Awareness – Cairo Monorail Project

Information Technology Institute (ITI): Developing Web Applications using PHP (2022)

Climathon 2022: Top 43 Teams – AWS & EU

Egypt University of Informatics: Top Achievers Award (2023, 2024)